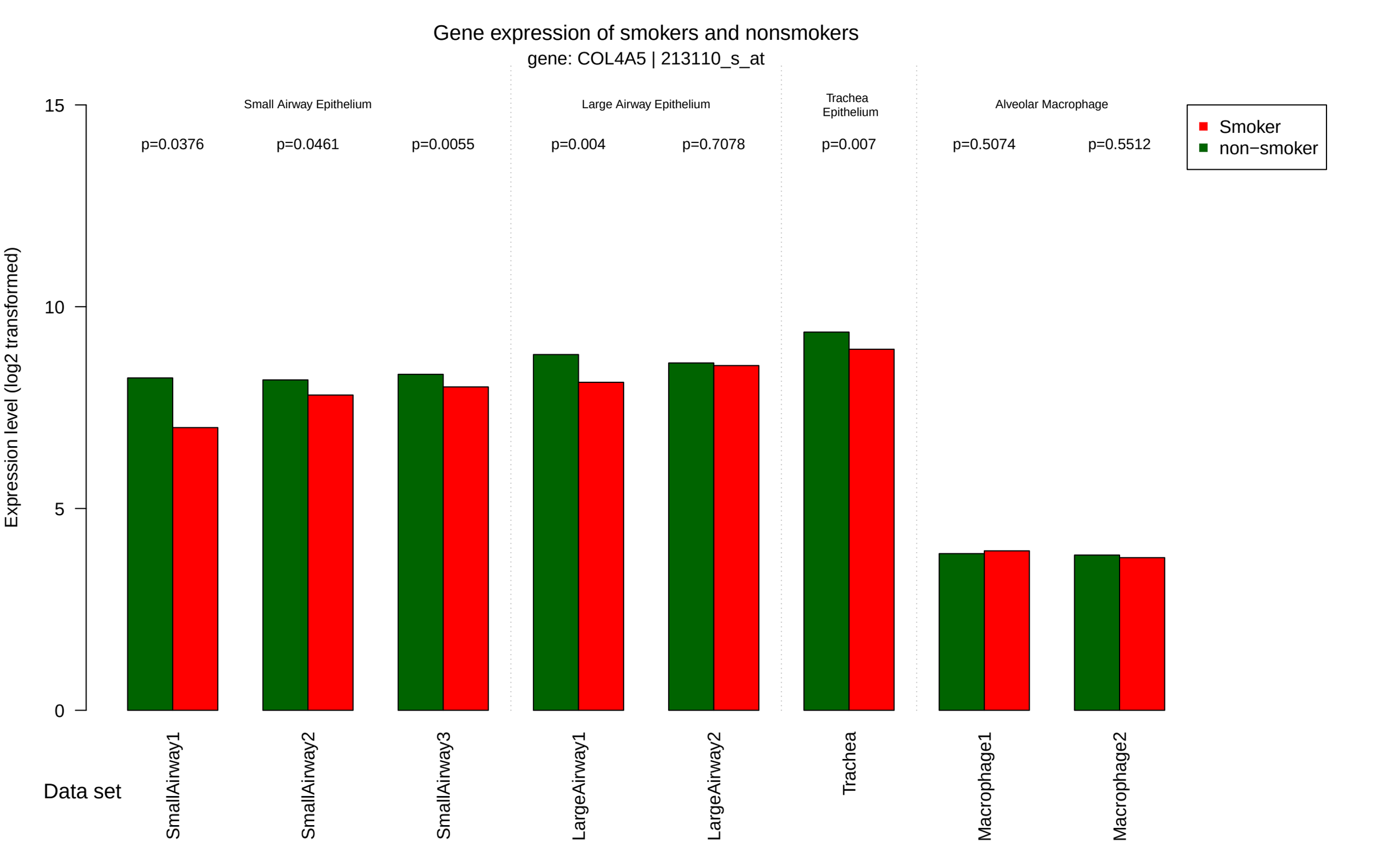
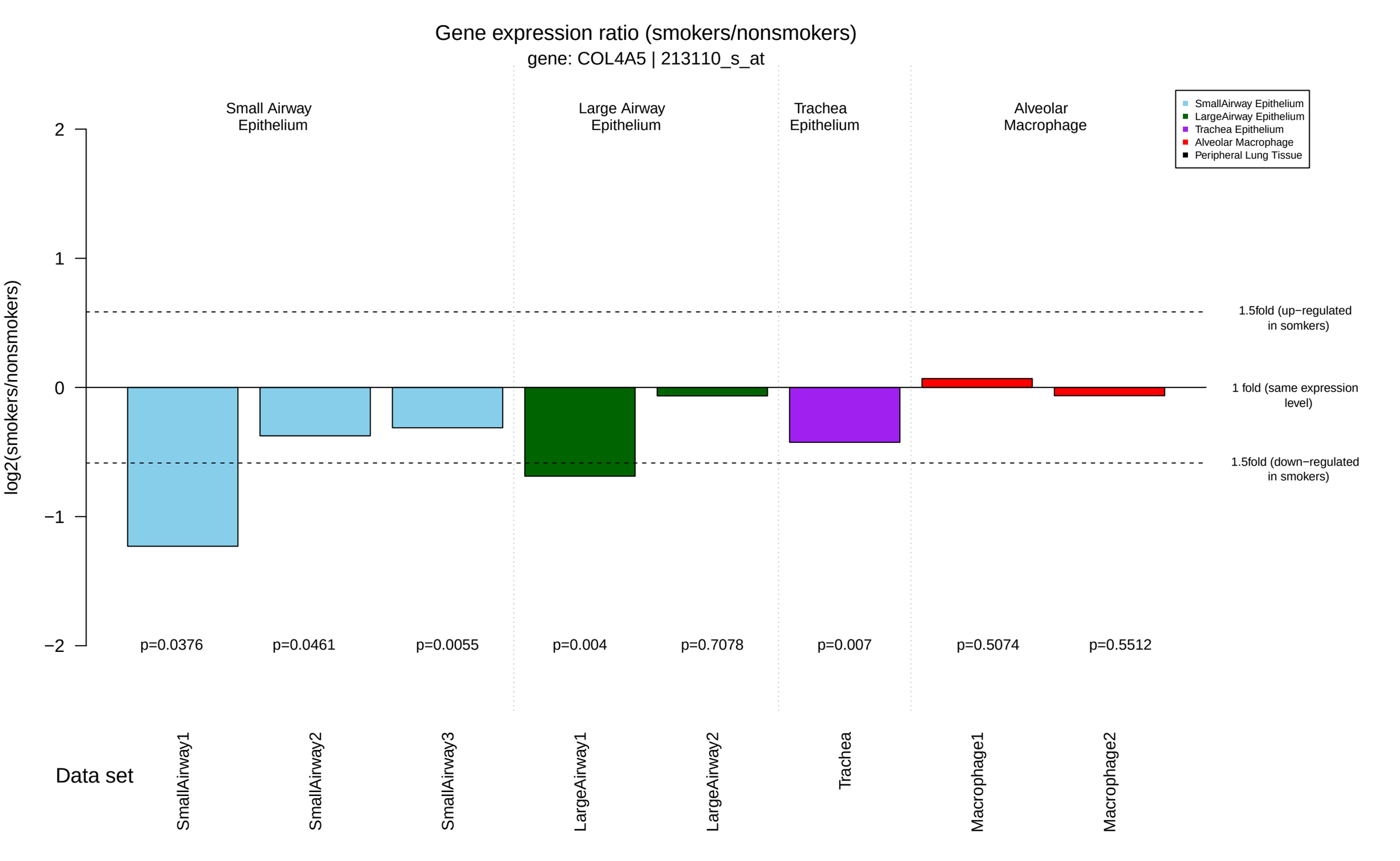
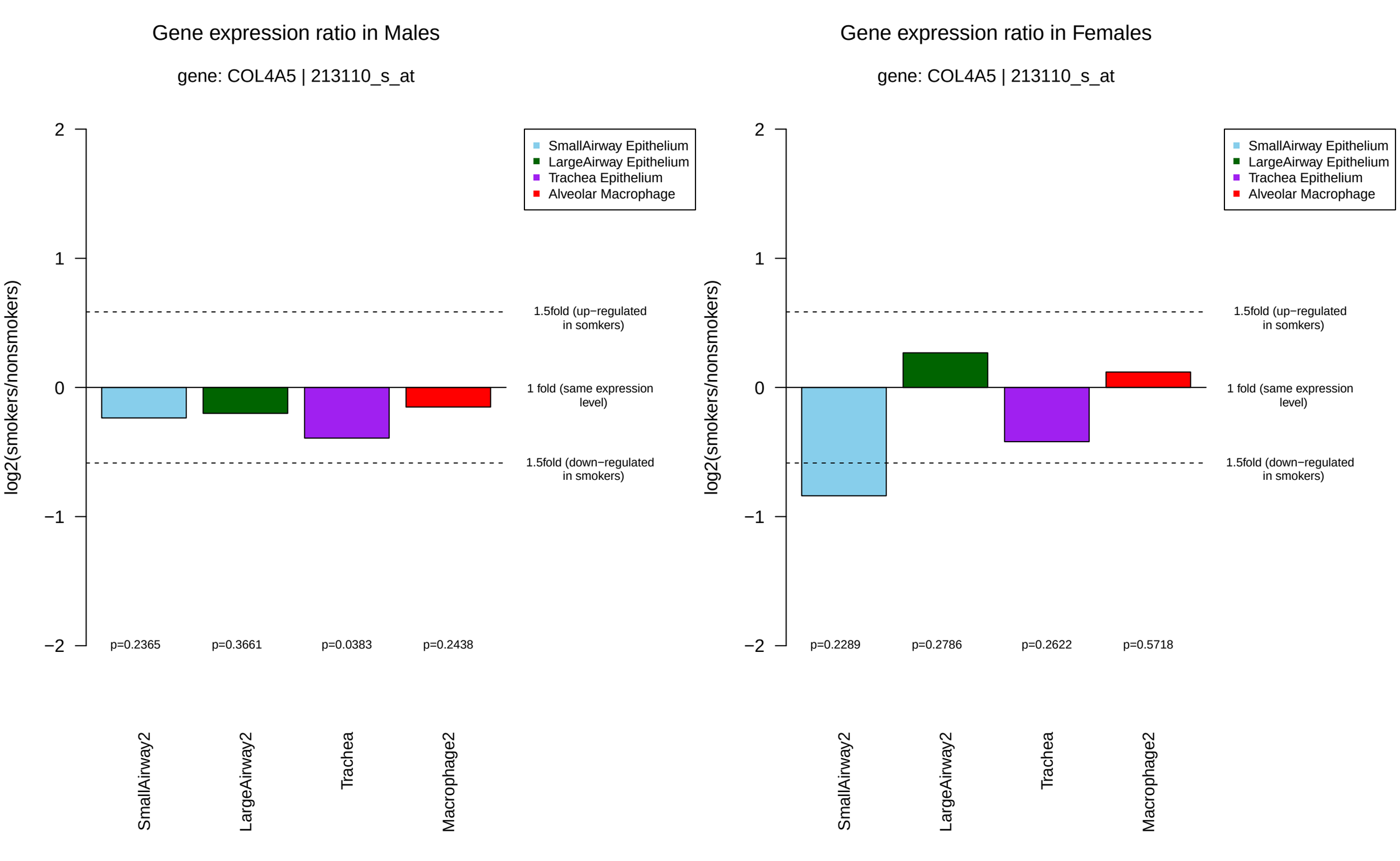
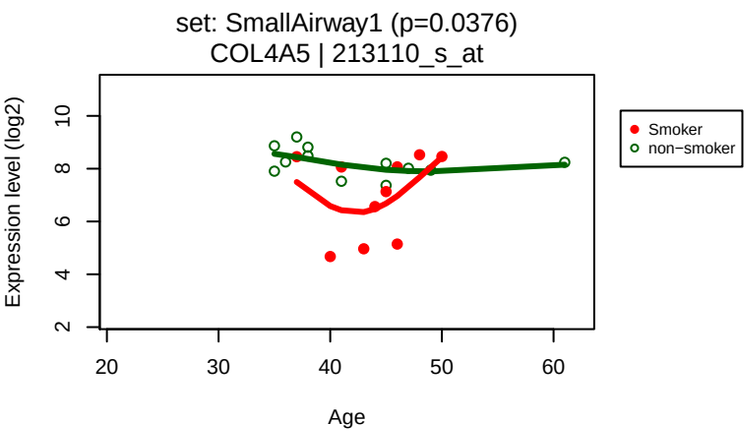
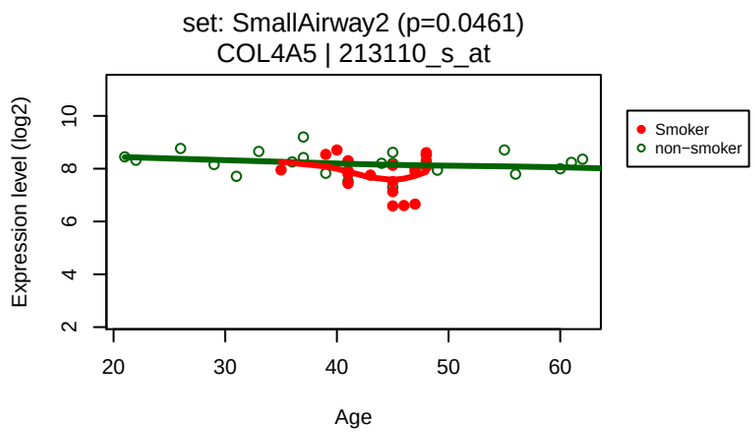
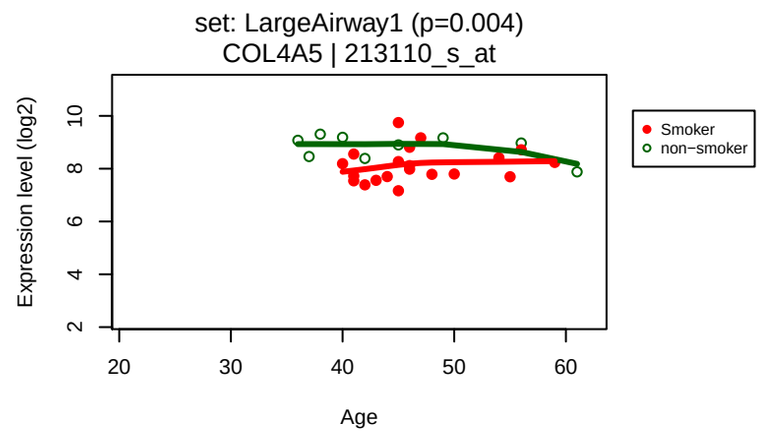

Gene expression of smokers and nonsmokers
gene: COL4A5 | 213110_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0376
p=0.0461
p=0.0055
p=0.004
p=0.7078
p=0.007
p=0.5074
p=0.5512
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

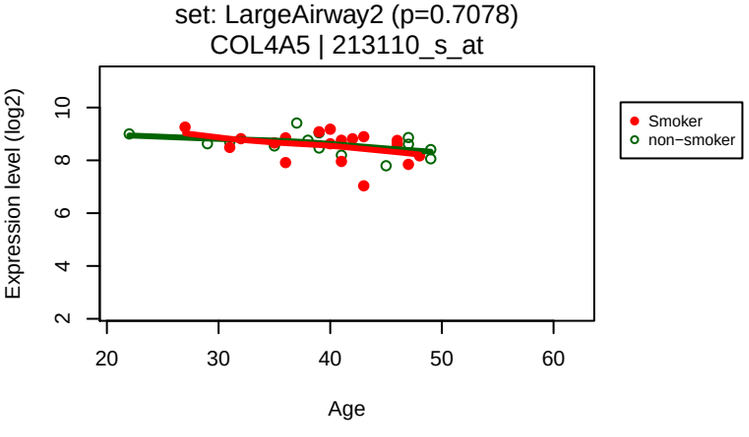

Gene expression ratio (smokers/nonsmokers)
gene: COL4A5 | 213110_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0376
p=0.0461
p=0.0055
p=0.004
p=0.7078
p=0.007
p=0.5074
p=0.5512
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

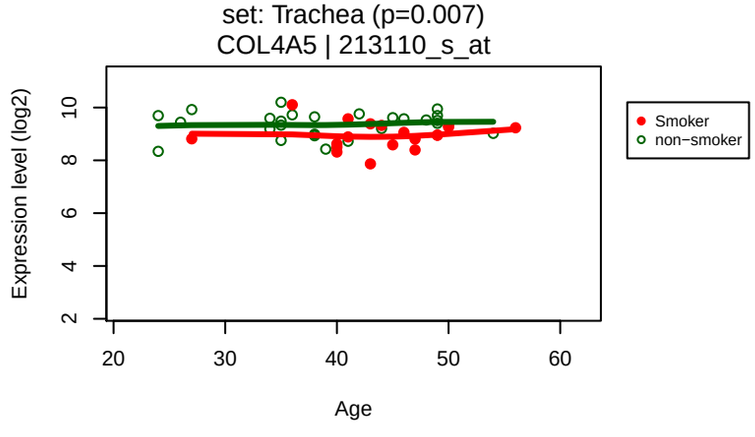

Gene expression ratio in Males
gene: COL4A5 | 213110_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2365
p=0.3661
p=0.0383
p=0.2438
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL4A5 | 213110_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2289
p=0.2786
p=0.2622
p=0.5718
SmallAirway2
LargeAirway2
Trachea
Macrophage2

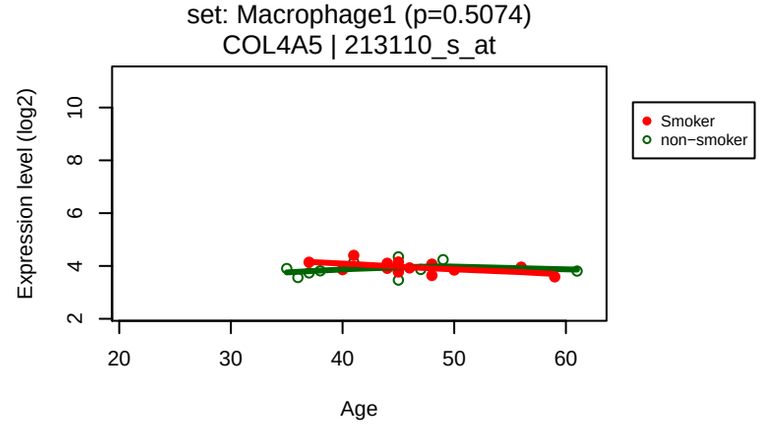

set: SmallAirway1 (p=0.0376)
COL4A5 | 213110_s_at
Expression level (log2)
Age
Smoker
non−smoker

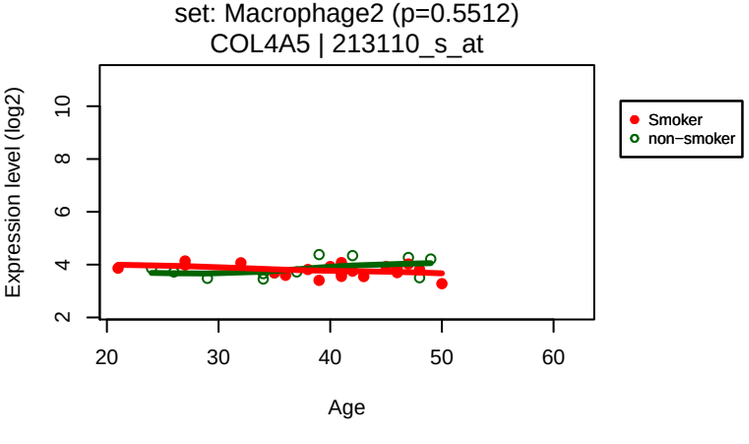

set: SmallAirway2 (p=0.0461)
COL4A5 | 213110_s_at
Expression level (log2)
Age
Smoker
non−smoker

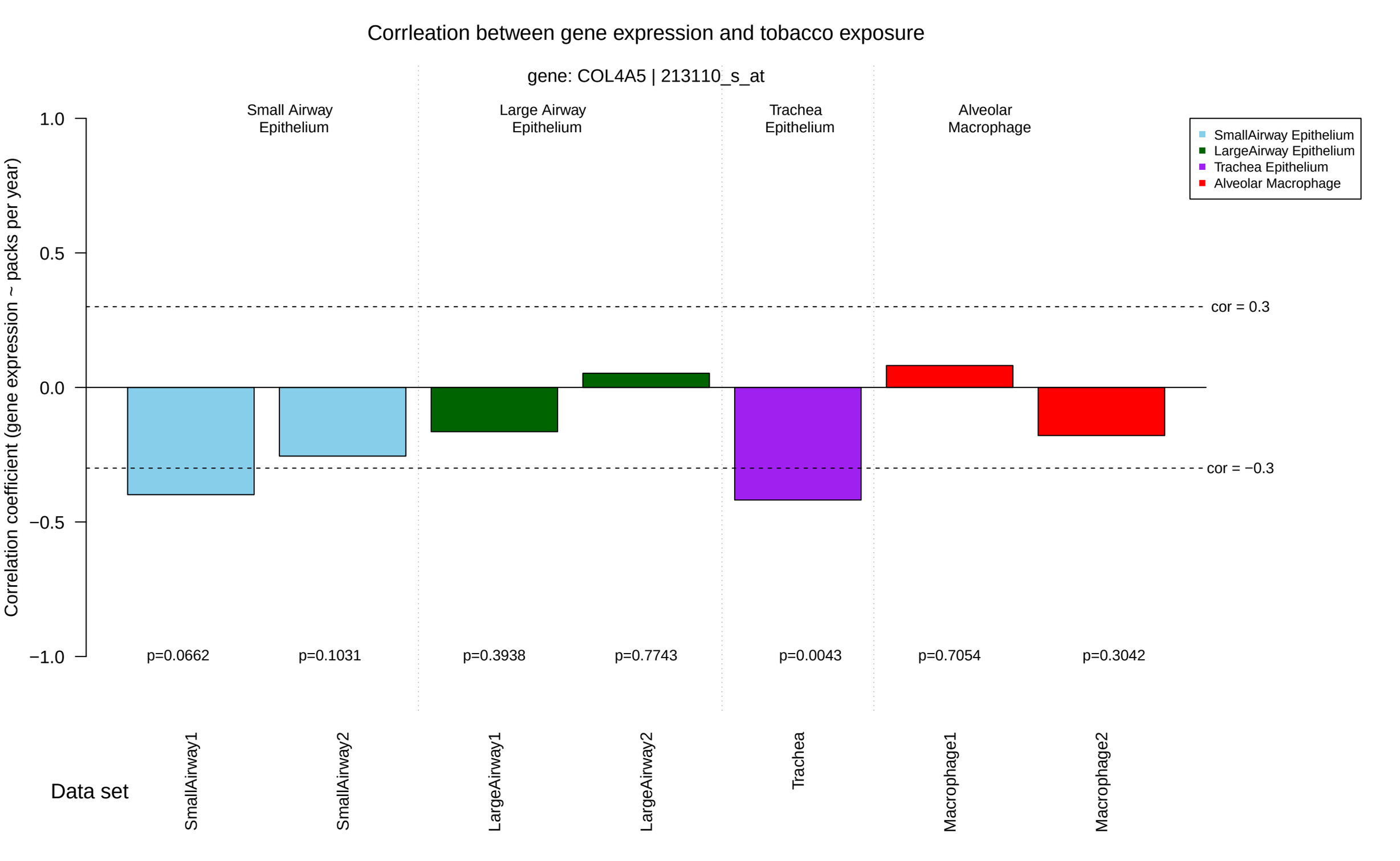

set: LargeAirway1 (p=0.004)
COL4A5 | 213110_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7078)
COL4A5 | 213110_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.007)
COL4A5 | 213110_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5074)
COL4A5 | 213110_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5512)
COL4A5 | 213110_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL4A5 | 213110_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0662
p=0.1031
p=0.3938
p=0.7743
p=0.0043
p=0.7054
p=0.3042
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2